

RETFound · Model A · Diabetic Retinopathy Grading

4-class ordinal grading (R0 · R1 · R2 · R3A) | Colour fundus photography | RETFound-DINOv2 ViT-L backbone

LINEAR PROBE (frozen backbone)

Only classification head trained · 50 epochs

0.841

Best Val AUROC

38

Best Val Epoch

Baseline — no test evaluation (interrupted at epoch 48, best ckpt saved)

+0.088

AUROC gain
LP → FT

FULL FINE-TUNE (all layers updated)

Layer-wise LR decay (0.65) · 50 epochs

0.929

Test AUROC

79.5%

Test Accuracy

0.639

Test κ

Best val epoch: 8 · Best val AUROC: 0.917

Dataset · Model A splits (images per class)

Split	R0	R1	R2	R3A	Total
Train	1942	1089	204	124	3359
Val	410	248	37	21	716
Test	391	236	51	24	702

Model & Training Setup

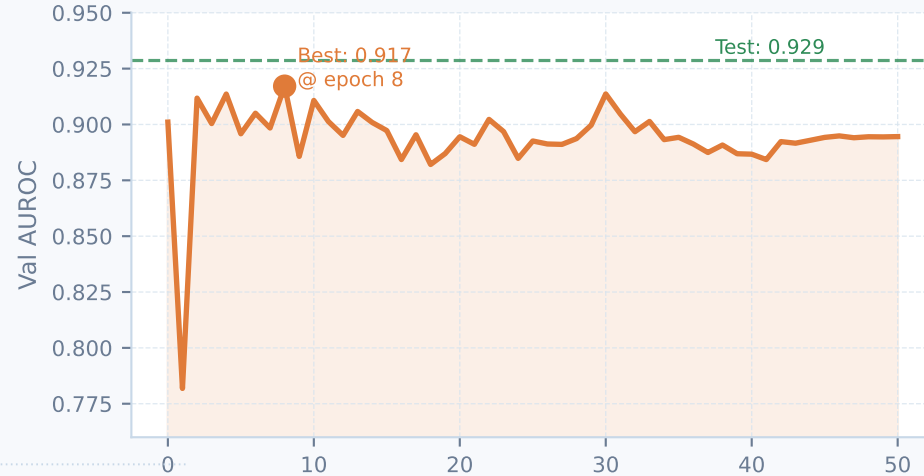
Backbone:	RETFound-DINOv2 ViT-Large (~307M params)
Pretrained on:	UK fundus cohort (MEH, 736k images)
Input size:	224 × 224 px
LP head:	Linear, frozen backbone · batch 64
FT LR decay:	Layer-wise × 0.65 per depth level
FT batch size:	24 (VRAM-safe for ViT-L)
Loss:	CrossEntropyLoss with class weights
Class weights:	R0=1.0 · R1=1.79 · R2=9.53 · R3A=15.68
Epochs:	50 for both LP and FT
Selection:	Best checkpoint by val AUROC

Training Dynamics — Validation Metrics over Epochs

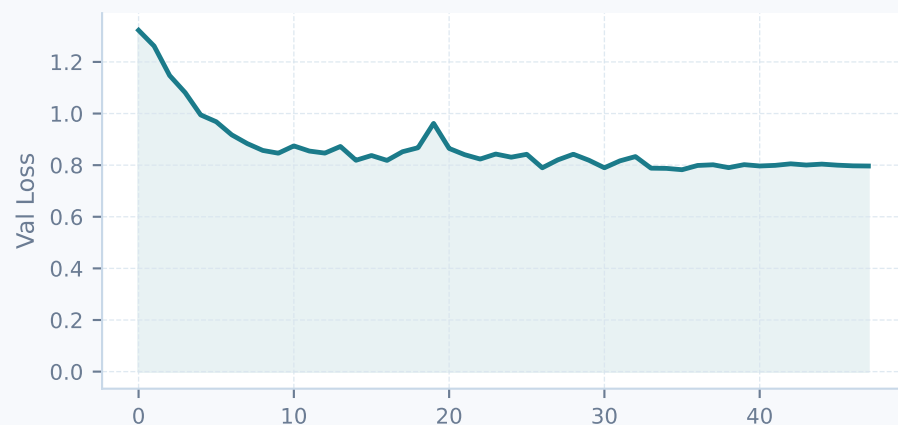
Linear Probe — Val AUROC



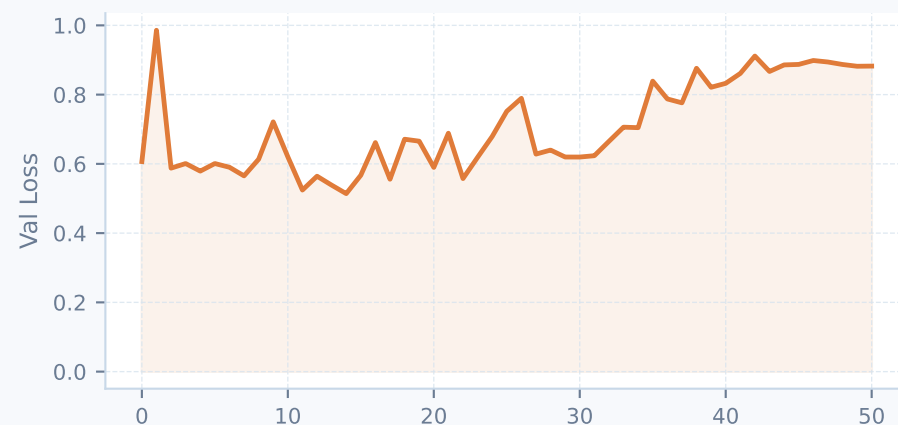
Full Fine-Tune — Val AUROC



Linear Probe — Val Loss



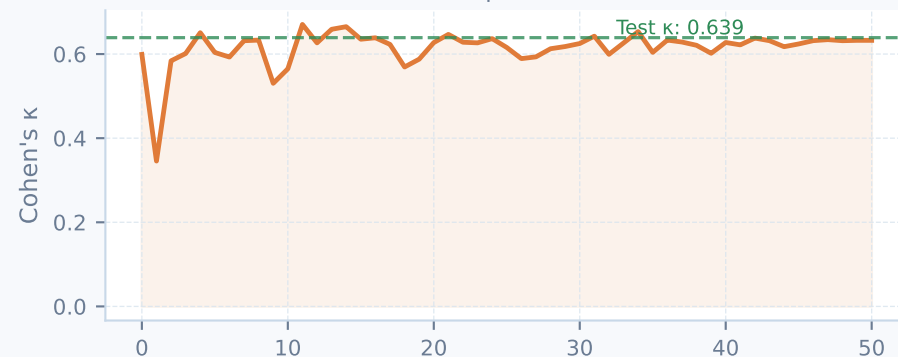
Full Fine-Tune — Val Loss



Linear Probe — Cohen's κ

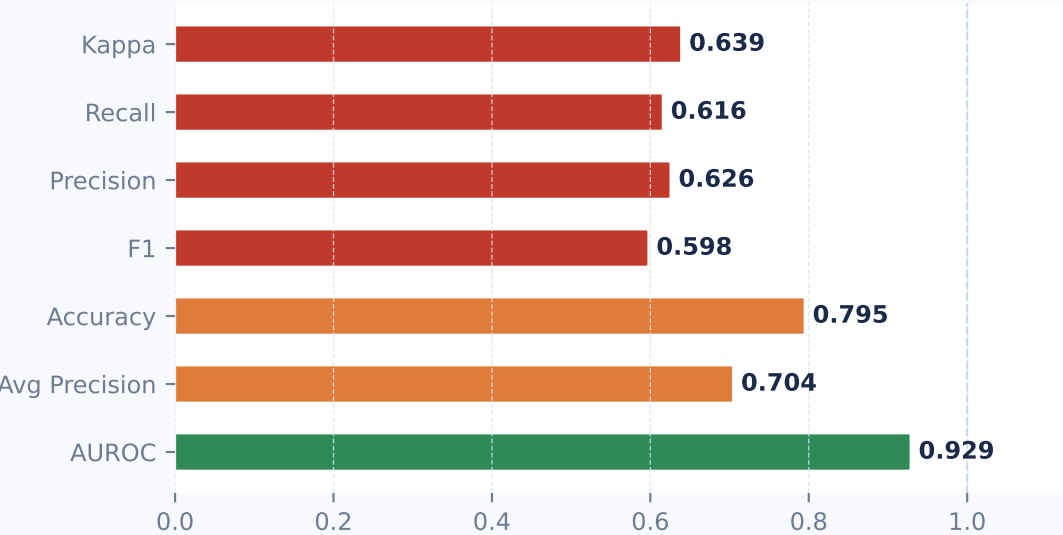


Full Fine-Tune — Cohen's κ



Test Set Evaluation — Full Fine-Tune (Best Checkpoint)

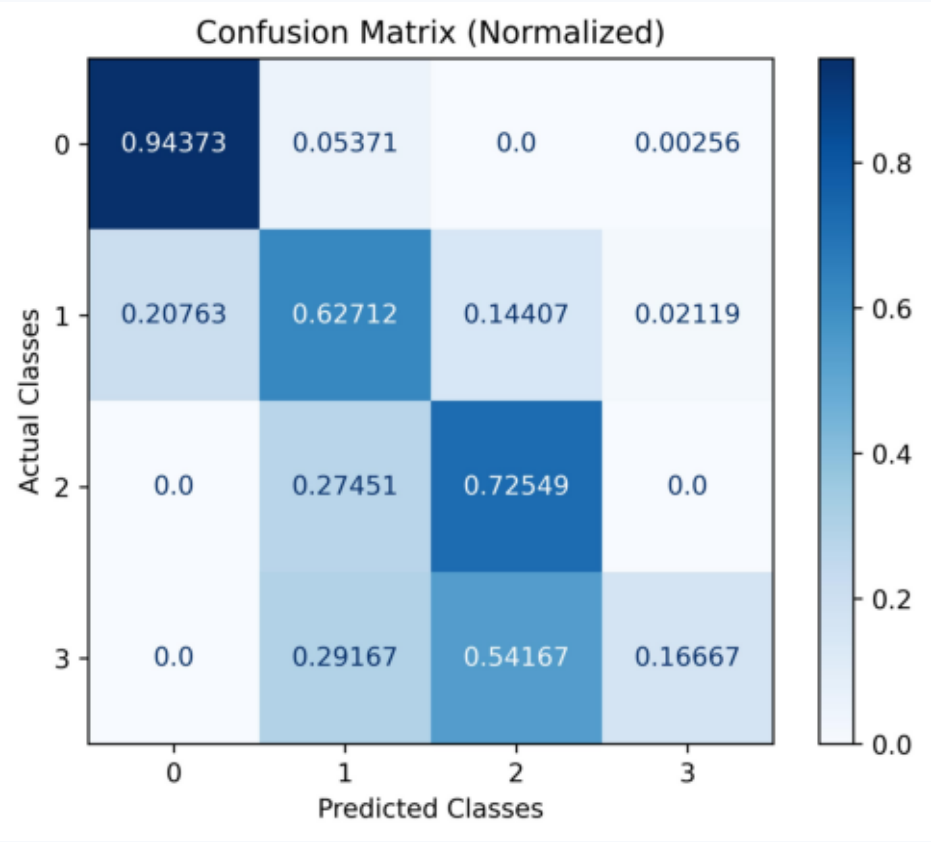
Test Set Metrics (Fine-Tune)



Linear Probe vs Full Fine-Tune — Key Metrics

Metric	Linear Probe (val, best ckpt)	Fine-Tune (test set)	Δ
AUROC	0.841	0.929	+0.088
Accuracy	~70%	79.5%	~+9.5pp
Kappa (κ)	~0.457	0.639	+0.182
F1	—	0.598	—
Precision	—	0.626	—
Recall	—	0.616	—
Avg Precision	—	0.704	—

Confusion Matrix — Test Set (Fine-Tune, Best Checkpoint)



AUROC 0.929 indicates strong class-separability across all 4 grades. $\kappa = 0.639$ (moderate-good) is clinically meaningful for a 4-class ordinal task. Lower F1 (0.60) reflects class imbalance: R2/R3A are rare and harder to recall.